

ChE 201 – Introduction to Thermodynamics –Fall 2015

Student Learning Outcome 1

Rubric for assessment problem on final exam (Problem #7)

Application of energy and entropy balances to a nozzle with ammonia flow

SLO 1 Performance Indicator (PI)	(1) Unsatisfactory	(2) Developing	(3) Satisfactory
1. Apply mathematical and scientific principles to formulate models and systems relevant to the subject Problem 7a, 7b, 7c 15 pts	Does not understand the connection between mathematical models and the system or process to be analyzed or designed (<7 pts)	Chooses a mathematical model or scientific principle that applies to an engineering problem, but has trouble in model development (7-9 pts)	Able to successfully combine mathematical and/or scientific principles to formulate models and systems relevant to chemical engineering (10-15 pts)
2. Solve engineering problems by using the concepts of algebra, calculus, differential equations and/or linear algebra Problem 7d, 7e 10pts	Does not understand the application of algebra, calculus, differential equations and/or linear algebra in solving chemical engineering problems (<4 pts)	Shows nearly complete understanding of applications of calculus and/or linear algebra in problem-solving (4-6 pts)	Applies concepts of algebra, calculus, differential equations and/or linear algebra to solve chemical engineering problems (7-10 pts)
4. Translates academic theory into engineering applications Problem 7f 5 pts	Does not appear to grasp the connection between theory and the problem (0-1 pts)	Some gaps in understanding the application of theory to the problem and expects theory to predict reality (2-3 pts)	Translates academic theory into engineering applications and accepts limitations of mathematical models of physical reality (4-5 pts)

Averages:

Performance Indicators: (1) 2.96 (2) 2.75 (4) 2.58

Overall Average: 2.76

Average scores showed 53/55 students were very strong in formulating the engineering system and model equations. A large majority were satisfactory or were developing the skills to solve engineering problems and translating theory into engineering applications (51/55 and 46/55).

QUESTION 7. (30 pts) Ammonia at 5 bar and 60°C , and a velocity of 10 m/s enters an insulated nozzle operating at steady state and exits at 2.5 bar and 20°C .

- a) (5 pts) Draw a schematic of the system and label the system boundary.
- b) (5 pts) State the engineering model
- c) (5 pts) Write the simplified energy balance and simplified entropy balance.
- d) (5 pts) Determine the velocity at the nozzle exit, in m/s
- e) (5 pts) Determine the entropy production in kJ/kg-K
- f) (5 pts) Sketch the process on an h-s diagram.

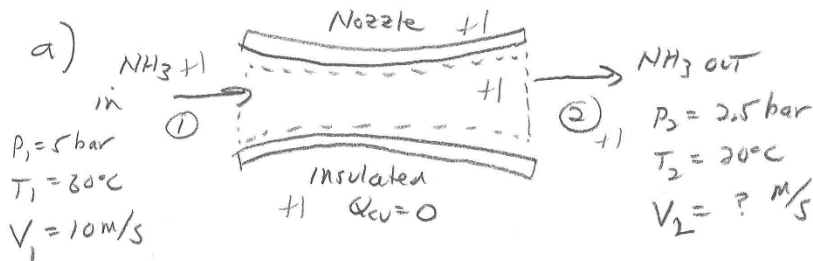
Solution

CHE 201 Final Exam

December 10, 2015

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- (5 pts) State the engineering model
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- 1) Nozzle is the control volume, steady state +1
- 2) $\dot{W}_{cv} = 0$, $Q_{cv} = 0$
- 3) $\Delta PE = 0$, ΔKE is significant
- 4) one inlet, one outlet

c) Energy Balance: (eqn 4.21) $0 = (h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2} \right)$

Entropy Balance: $\Delta S = \int_1^2 \left(\frac{\delta Q}{T} \right)_b + \sigma_{cv}$

$(s_2 - s_1) = \sigma_{cv} \left(\frac{\text{kJ}}{\text{kg-K}} \right)$

d) $V_2 = ?$ Rearrange E.B. in terms of V_2 :

$$\frac{V_2^2}{2} = h_1 - h_2 + \frac{V_1^2}{2} \Rightarrow V_2 = \left(2(h_1 - h_2) + V_1^2 \right)^{1/2} + 1$$

$V_1 = 10 \text{ m/s}$ Table A-15 $h_1 = 1585.81 \frac{\text{kJ}}{\text{kg}}$ $5 \text{ bar} = P_1$ $+1$
 $60^\circ\text{C} = T_1$

$h_2 = 1505.8 \frac{\text{kJ}}{\text{kg}}$ $2.5 \text{ bar} = P_2$ $+1$
 $20^\circ\text{C} = T_2$

$$+1 \quad V_2 = \left\{ 2 \left(1585.81 - 1505.8 \right) \frac{\text{kJ}}{\text{kg}} \left| \frac{1 \text{ kg} \cdot \text{m/s}^2}{1 \text{ N}} \right| \frac{10^3 \text{ N} \cdot \text{m}}{1 \text{ kJ}} \right\}^{1/2} + \left(10 \frac{\text{m}}{\text{s}} \right)^2 \left\{ \right. \\ \left. = \left\{ 2(80.01)(1000) \frac{\text{m}^2}{\text{s}^2} + 100 \frac{\text{m}^2}{\text{s}^2} \right\}^{1/2} \right. \\ \left. \left\{ 160,020 + 100 \right\} \frac{\text{m}^2}{\text{s}^2} \right\}^{1/2} = \left(160,120 \frac{\text{m}^2}{\text{s}^2} \right)^{1/2} + 1$$

$$V_2 = 400.15 \text{ m/s}$$

e) $\sigma_{cv} = (s_2 - s_1)$

Table A-15 $s_1 = 5.7382 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$
 $+P \quad s_2 = 5.8083$

$$= (5.8083 - 5.7382) \frac{\text{kJ}}{\text{kg} \cdot \text{K}} + P$$

$+1 \quad \sigma_{cv} = +0.0721 \frac{\text{kJ}}{\text{kg} \cdot \text{K}}$

f)

